

- Long Papers*), pages 517–539, Mexico City, Mexico, June 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.naacl-long.29. URL <https://aclanthology.org/2024.naacl-long.29>.
- Ivan Sekulic, Silvia Terragni, Victor Guimarães, Nghia Khau, Bruna Guedes, Modestas Filipavicius, Andre Ferreira Manso, and Roland Mathis. Reliable LLM-based user simulator for task-oriented dialogue systems. In Yvette Graham, Qun Liu, Gerasimos Lampouras, Ignacio Iacobacci, Sinead Madden, Haider Khalid, and Rameez Qureshi, editors, *Proceedings of the 1st Workshop on Simulating Conversational Intelligence in Chat (SCI-CHAT 2024)*, pages 19–35, St. Julians, Malta, March 2024. Association for Computational Linguistics. URL <https://aclanthology.org/2024.scichat-1.3>.
- Sailik Sengupta, Zahra Zahedi, and Subbarao Kambhampati. To monitor or to trust: observing robot’s behavior based on a game-theoretic model of trust. In *Proceedings of the Trust Workshop at AAMAS*, 2019.
- AB Siddique, Muhammad Hasan Maqbool, Kshitija Taywade, and Hassan Foroosh. Personalizing task-oriented dialog systems via zero-shot generalizable reward function. In *Proceedings of the 31st ACM International Conference on Information & Knowledge Management*, pages 1787–1797, 2022.
- Zhongyu Wei, Qianlong Liu, Baolin Peng, Huaixiao Tou, Ting Chen, Xuan-Jing Huang, Kam-Fai Wong, and Xiang Dai. Task-oriented dialogue system for automatic diagnosis. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 201–207, 2018.
- Bosi Wen, Pei Ke, Xiaotao Gu, Lindong Wu, Hao Huang, Jinfeng Zhou, Wenchuang Li, Binxin Hu, Wendy Gao, Jiaxin Xu, Yiming Liu, Jie Tang, Hongning Wang, and Minlie Huang. Benchmarking complex instruction-following with multiple constraints composition, 2024. URL <https://arxiv.org/abs/2407.03978>.
- Dingjun Wu, Jing Zhang, and Xinmei Huang. Chain of thought prompting elicits knowledge augmentation. In Anna Rogers, Jordan Boyd-Graber, and Naoaki Okazaki, editors, *Findings of the Association for Computational Linguistics: ACL 2023*, pages 6519–6534, Toronto, Canada, July 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.findings-acl.408. URL <https://aclanthology.org/2023.findings-acl.408>.
- Lin Xu, Ningxin Peng, Daquan Zhou, See-Kiong Ng, and Jinlan Fu. Chain of thought explanation for dialogue state tracking. *arXiv preprint arXiv:2403.04656*, 2024.
- An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, Chujie Zheng, Dayiheng Liu, Fan Zhou, Fei Huang, Feng Hu, Hao Ge, Haoran Wei, Huan Lin, Jialong Tang, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiayi Yang, Jing Zhou, Jingren Zhou, Junyang Lin, Kai Dang, Keqin Bao, Kexin Yang, Le Yu, Lianghao Deng, Mei Li, Mingfeng Xue, Mingze Li, Pei Zhang, Peng Wang, Qin Zhu, Rui Men, Ruize Gao, Shixuan Liu, Shuang Luo, Tianhao Li, Tianyi Tang, Wenbiao Yin, Xingzhang Ren, Xinyu Wang, Xinyu Zhang, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yinger Zhang, Yu Wan, Yuqiong Liu, Zekun Wang, Zeyu Cui, Zhenru Zhang, Zhipeng Zhou, and Zihan Qiu. Qwen3 technical report, 2025. URL <https://arxiv.org/abs/2505.09388>.

Appendix

A Datasets

In this section, we present sample data from our proposed TOD-ProcBench and related works.

SGD (Task: Flights_1)

description: Find your next flight
slots: [passengers, seating_class, origin_city, destination_city, origin_airport, ...]
slot: seating class
description: Seating class for the booking
is_categorical: true
possible_values: [Economy, Premium Economy, Business, First Class]
intents: [SearchOnewayFlight, SearchRoundtripFlights, ReserveOnewayFlight, ReserveRoundtripFlights]
intent: ReserveOnewayFlight
description: Reserve a one-way flight
is_transactional: True
required_slots: [origin_city, destination_city, airlines, departure_date]
optional_slots: [refundable, passengers, seating_class]

Table 5: Example of instructions in SGD.

STAR (Task: book_doctor_appointment)

graph: Hello → ask_name → ask_doctor_name → ask_day → ask_start_time → ask_symptoms → query_check
available → inform_booking_available
unavailable → inform_booking_unavailable
yes → query_book
no → ask_doctor_name
query_book → doctor_inform_booking_successful → inform_booking_successful

Table 6: Example of instructions in STAR.

ABCD (Task: Policy and Return Due to Size)

search-faq → search-policy → select-faq
pull-up-account → validate-purchase → membership → enter-details → update-order

Table 7: Example of instructions in ABCD.

TOD-ProcBench (Task: Policy and Return Due to Size)

- If customer wants to know the return policy:
 - Ask for customer's membership level (gold, silver, bronze, guest)
 - If gold:
 - Allow unlimited returns
 - If silver:
 - Allow returns for purchases made in the last 6 months
 - If bronze:
 - Allow returns for purchases made in the last 90 days
 - If guest:
 - Allow returns for purchases made in the last 30 days
- If customer wants to return an item due to wrong size:
 - Ask for customer's full name or account ID
 - Ask for username, email address, and order ID to validate purchase
 - Ask for membership level (guest, bronze, silver, gold)
 - If membership is guest:
 - If purchase was within last 30 days or customer has receipt/original packaging:
 - Allow return
 - Else:
 - Deny return
 - If membership is bronze:
 - If purchase was within last 90 days or customer has receipt/original packaging:
 - Allow return
 - Else:
 - Deny return
 - If membership is silver:
 - If purchase was within last 6 months or customer has receipt/original packaging:
 - Allow return
 - Else:
 - Deny return
 - If membership is gold:
 - Allow unlimited returns
 - If return is allowed:
 - Ask for customer's full address to generate shipping label
 - Ask how customer wants to process return (by mail, in store, drop off center)
 - Provide shipping label and return instructions

Table 8: Example of instructions in TOD-ProcBench.